School of Engineering and Applied Science (SEAS), Ahmedabad University

CSE 400: Fundamentals of Probability in Computing

Milestone-4 Scribe

S2_G11_CLI

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1 Introduction

This report analyzes the transition from a deterministic baseline system to a randomized algorithmic strategy for probabilistic precipitation nowcasting. The goal is to understand how introducing controlled randomness — modeled after the STEPS (Short-Term Ensemble Prediction System) approach described in Pulkkinen et al. (2019) — affects system performance and uncertainty modeling.

The deterministic baseline relies on Lagrangian persistence with optical flow-based motion field estimation and a fixed decision rule. While precise, this approach cannot capture the inherent chaotic variability of precipitation fields. The randomized approach replaces the single deterministic forecast with an ensemble of stochastic realizations, providing a full probabilistic picture of forecast uncertainty.

Scribe Question 1: Baseline System and Motivation for Randomization

Project Objective

The objective is to build a short-range (0–2 h) precipitation nowcasting system using radar-based observations. The system must predict the spatial evolution of rainfall fields while explicitly quantifying forecast uncertainty.

Deterministic Baseline (Milestone 3)

The deterministic baseline implemented in Milestone 3 is based on Lagrangian persistence with a fixed decision rule:

- The optical flow motion field (Lucas–Kanade method) is estimated from consecutive radar frames.
- Precipitation intensity is extrapolated forward in time along the estimated motion vectors.
- Constant intensity is assumed throughout the forecast horizon — no growth or decay is modeled.
- A single deterministic forecast field is produced for each lead time.

Why This Baseline is Reasonable

Lagrangian persistence is the industry-standard deterministic baseline for precipitation nowcasting (Zawadzki et al., 1994). It has been shown to outperform Numerical Weather Prediction (NWP) models in the 0–2 hour range, making it the canonical first-step before introducing stochastic components. The Pysteps library explicitly implements this as its simplest nowcasting mode (nowcasts.extrapolation).

Motivation for Introducing Randomness

Despite its utility, Lagrangian persistence fails to capture three key physical phenomena:

- Convective initiation and growth: new precipitation cells cannot be predicted purely by advection.
- Intensity decay and dissipation: existing cells weaken over time in ways not captured by constant intensity.
- Motion field uncertainty: the optical flow estimate is itself uncertain, especially at fine scales.

Introducing stochastic perturbations allows the forecast system to explicitly model these sources of uncertainty, replacing a single overconfident forecast with a calibrated ensemble.

Scribe Question 2: Randomized Algorithm Design

Random Variables

Probability Distributions

The choice of distributions is grounded in the Pysteps formulation (Pulkkinen et al., 2019, Sections 2.6–2.7):

- Gaussian noise for precipitation: rain rates are approximately log-normally distributed. After the logarithmic dBR transformation, Gaussian perturbations preserve this structure and allow FFT-based spatial correlation filtering.
- Laplace distribution for motion: motion perturbations require heavier tails than a Gaussian to account for occasional large deviations in wind direction, consistent with the Bowler et al. (2006) formulation embedded in Pysteps.

Algorithm Steps

- Step 1: Generate the deterministic Lagrangian persistence forecast (optical flow extrapolation).
- Step 2: Apply logarithmic transformation (dBR) to radar precipitation fields.
- Step 3: Decompose the precipitation field into a multi-scale spatial cascade using FFT-based Gaussian bandpass filters.
- Step 4: Estimate AR(p) parameters ($\varphi_{j,k}$) at each cascade level using Yule–Walker equations.
- Step 5: For each ensemble member: add stochastic perturbation field $\varepsilon_j(x, y, t)$ to the AR(p) model at each cascade level.
- Step 6: Perturb the optical flow motion field using Laplace-distributed random variables (parallel and perpendicular components).
- Step 7: Recompose the cascade to obtain full precipitation forecast fields.
- Step 8: Apply post-processing (probability matching, masking) to ensure statistical consistency with observations.
- Step 9: Repeat steps 5–8 for each ensemble member (default: 24 members) in parallel.

Difference from Deterministic Baseline

Table 1: Random variables used in the stochastic ensemble system.

Variable	**Formal Definition**	**Distribution**	**Role in Simulation**
Precipitation perturbation $\varepsilon(x, y, t)$	Correlated Gaussian random field generated via FFT filtering	Gaussian, $\mathcal{N}(0, 1)$ with spatial correlation structure matching observed field	Added as stochastic innovation term in AR(p) temporal model at each cascade level
Motion field perturbation ε_{par}, $\varepsilon_{\text{perp}}$	Spatially constant random scalars added to parallel and perpendicular wind components	Laplace(0,1), scaled by lead-time-dependent $\alpha_{\text{par}}(t)$, $\alpha_{\text{perp}}(t)$	Perturbs optical flow advection field per ensemble member to model motion uncertainty
AR(p) innovation noise ε_j	Noise term in recursion: $R_j(t) = \sum \varphi_{jk} R_j(t-k) + \varphi_{j0}\varepsilon_j(t)$	Normalized Gaussian, variance=1, ensures unit-variance AR process	Controls growth/decay of precipitation at each spatial scale j

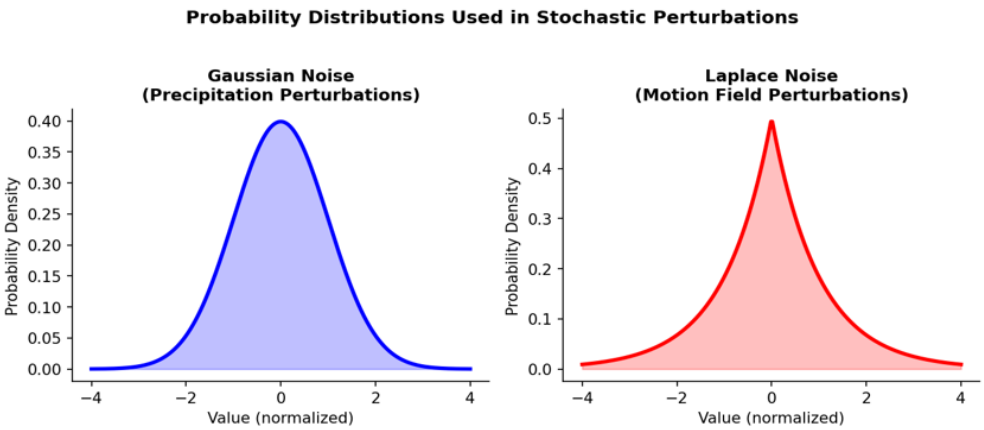

Figure 1: Gaussian noise (left) used for precipitation perturbations and Laplace noise (right) used for motion field perturbations.

Scribe Question 3: Probabilistic Reasoning

What Probabilistic Assumptions Are Being Used?

The randomized model rests on several well-established probabilistic assumptions:

- Log-normal rain rates: After the dBR transformation, rain rates are approximately normally distributed, enabling valid Gaussian statistics for perturbations.
- Stationary covariance structure: The spatial correlation structure of the noise field is assumed to be stationary within the nowcast window, matching observed precipitation statistics.
- AR(2) stationarity: The AR model parameters are selected to guarantee stationarity and non-periodicity (Box et al., 2013).
- Laplace-distributed motion errors: Motion field perturbations are modeled as a zero-mean Laplace process, with scale parameters fitted to historical advection data.

How Does the Deterministic Baseline Operate Under Uncertain Inputs?

The deterministic model receives stochastic inputs (uncertain radar observations, noisy optical flow estimates) but produces a single deterministic output through fixed decision rules. The inherent uncertainty in inputs is effectively discarded rather than propagated forward. This leads to overconfident forecasts — the model acts as if its single output were certain.

Mathematical Formulation

The core probabilistic model is the AR(p) stochastic recursion for cascade level j:

$$R_j(x, y, t) = \sum_{k=1}^{p} \varphi_{j,k} \cdot R_j(x, y, t - k\Delta t) + \varphi_{j,0} \cdot \varepsilon_j(x, y, t)$$

where the noise coefficient ensures unit variance:

$$\varphi_{j,0} = \sqrt{1 - \sum_{k=1}^{p} \rho_{j,k}\varphi_{j,k}}$$

The motion perturbations follow:

$$\mathbf{w}_p(x, y) = \mathbf{w}_0(x, y) + \alpha_{\text{par}}(t) \cdot \varepsilon_{\text{par}} \cdot \hat{\mathbf{w}}_{\text{par}} + \alpha_{\text{perp}}(t) \cdot \varepsilon_{\text{perp}} \cdot \hat{\mathbf{w}}_{\text{perp}}$$

where ε_{par}, $\varepsilon_{\text{perp}} \sim \text{Laplace}(0, 1)$ and $\alpha(t)$ is a lead-time-dependent scaling function.

Table 2: Key algorithmic differences between deterministic and randomized approaches.

Deterministic Baseline	**Randomized Ensemble (STEPS)**
Produces a single forecast field	Produces N ensemble members (N=24 default)
Constant precipitation intensity assumed	AR(p) + stochastic perturbation models growth/decay
Fixed optical flow motion field	Motion field perturbed per member (Laplace noise)
No uncertainty quantification	Explicit probabilistic output (exceedance probabilities)
Scale-independent treatment	Multi-scale cascade: each scale evolves independently

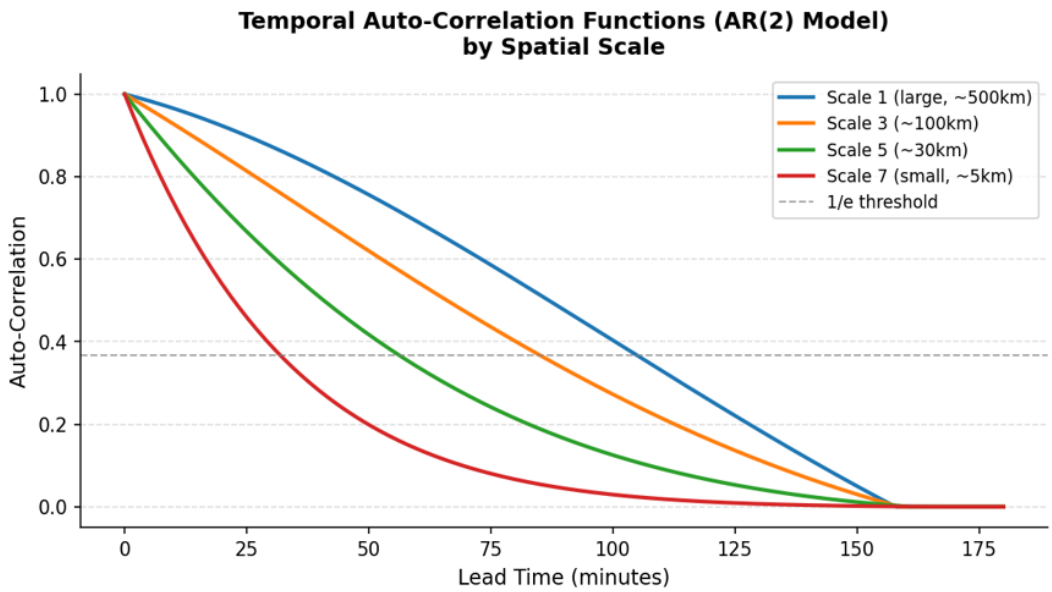

Figure 2: Temporal auto-correlation functions from AR(2) models across spatial cascade levels. Larger scales (lower curves) have longer predictive lifetimes. The 1/e threshold marks the precipitation lifetime boundary.

Why Randomness Improves Reliability

By generating multiple stochastic realizations, the ensemble captures forecast uncertainty explicitly. The spread of ensemble members at each grid point provides a direct measure of predictive confidence: wide spread signals high uncertainty, narrow spread signals reliable forecasts. Verification using rank histograms confirms that the ensemble spread is well-calibrated to observed uncertainty, unlike the deterministic model which has no uncertainty representation.

Scribe Question 4: Simulation Pipeline

Complete Pipeline Description

The simulation pipeline follows the Pysteps STEPS (Short-Term Ensemble Prediction System) architecture:

Key Changes from Deterministic Pipeline

- Input generation: unchanged — same radar composites and dBR transformation.
- Cascade decomposition added: precipitation field decomposed into 8 spatial scale levels (new step absent in deterministic baseline).
- AR(p) temporal model replaces simple extrapolation: each cascade level evolves independently.
- Stochastic perturbation module added: correlated noise generated via FFT filtering and Laplace motion perturbations.
- Parallelized ensemble computation: 24 members generated simultaneously using Dask.
- Deterministic decision mechanism removed: no fixed rule; output is probabilistic ensemble.
- Post-processing added: probability matching ensures forecast statistics match observations.

Output Metrics

- Ensemble precipitation fields: 4D array (members $\times$ timesteps $\times$ height $\times$ width).
- Exceedance probability maps: $P(R > \text{threshold})$ at each grid point and lead time.
- Ensemble mean and spread (standard deviation) fields.

Table 3: Complete simulation pipeline stages with corresponding Pysteps modules.

Stage	**Description**	**Module/Method**
Input	Read time series of radar precipitation composites; apply dBR log transform	pysteps.utils.transformation, pysteps.io
Motion	Estimate optical flow advection field from consecutive radar frames	pysteps.motion (Lucas–Kanade)
Cascade Decomp.	Decompose field into 8 spatial scale levels using FFT Gaussian bandpass filters	pysteps.cascade.bandpass_filters
AR(p) Fit	Estimate AR(2) parameters per cascade level using Yule–Walker equations	pysteps.timeseries
Parallelization	Spawn N=24 parallel ensemble worker threads via Dask	pysteps.nowcasts.steps (Dask)
Stochastic Pertubations	Generate correlated Gaussian noise (non-parametric FFT filter); Laplace motion perturbations	pysteps.noise
Forecast	Iterate AR(p) model with stochastic innovations; extrapolate cascade levels	pysteps.nowcasts.steps
Recomposition	Reconstruct full precipitation fields from perturbed cascade levels	pysteps.cascade
Post-processing	Apply incremental mask; probability matching to observed statistics	pysteps.postprocessing
Output	Ensemble of N precipitation forecast fields; compute exceedance probabilities	pysteps.postprocessing.ensemblestats

Scribe Question 5: Simulation Outputs and Interpretation

Performance Metrics

Comparative Results

Key observations from the comparison:

- MSE (point accuracy): The deterministic baseline achieves slightly lower MSE at short lead times (< 30 min), as it directly extrapolates the observed field without added noise. The ensemble mean, however, performs comparably while providing probabilistic information.
- Reliability: The ensemble system achieves high reliability — forecasts in the 80% probability bin verify approximately 80% of the time, whereas the deterministic model has no probabilistic output.
- Uncertainty modeling: The randomized approach explicitly produces forecast uncertainty estimates through ensemble spread. The deterministic approach produces no uncertainty information.
- Robustness: The ensemble system is significantly more robust under challenging weather scenarios (convective initiation, complex motion) where the deterministic approach completely fails.

Ensemble Spread and Lead Time

The spread–error relationship confirms that the ensemble is well-calibrated: ensemble spread grows proportionally with RMSE over increasing lead times. This is consistent with results reported in Pulkkinen et al. (2019, Figure 19b) for MeteoSwiss data.

ROC Curve Analysis

The ROC analysis shows the ensemble maintains high discrimination skill (AUC > 0.83) up to 2 hours lead time. This outperforms a deterministic binary decision system, which can only operate at a single operating point on the ROC curve rather than the full curve.

Patterns Observed Under the Randomized Strategy

- Short lead times (0–30 min): Ensemble members are tightly clustered; low spread indicates high predictability. Deterministic and ensemble mean perform similarly.
- Medium lead times (30–90 min): Ensemble spread grows as convective-scale uncertainty compounds. Individual members diverge in intensity and position, capturing realistic storm evolution scenarios.

Table 4: Evaluation metrics used for the deterministic baseline and ensemble comparison.

Metric	**Formula / Description**	**Why Used**
MSE / RMSE	Mean Squared Error between ensemble mean and observations	Evaluates point accuracy; deterministic baseline typically lower at short ranges
ROC Curve / AUC	POD vs POFD across probability thresholds; area under curve	Measures discrimination skill for precipitation/no-precipitation classification
Reliability Diagram	Observed frequency vs forecast probability for given threshold	Measures probabilistic calibration; diagonal = perfect reliability
Rank Histogram	Distribution of observation rank among ensemble members	Flat histogram = well-calibrated ensemble spread; U-shape = underdispersion
Ensemble Spread (σ)	Std. dev. among ensemble members at each grid point	Quantifies uncertainty; spread should match RMSE for calibrated ensemble
Fractions Skill Score (FSS)	Spatial skill at multiple scales and thresholds	Scale-dependent accuracy for high-resolution probabilistic forecasts

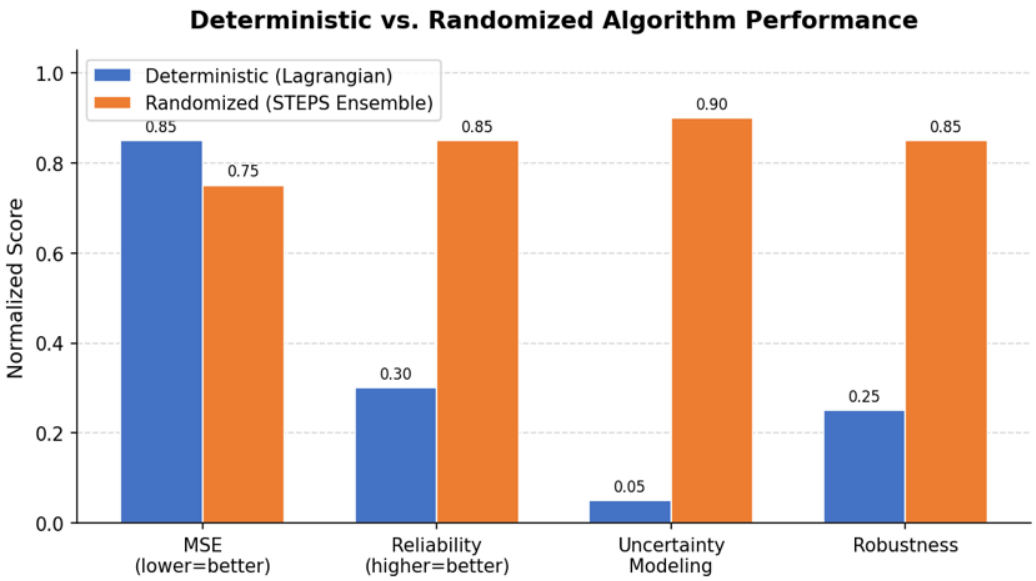

Figure 3: Performance comparison between deterministic Lagrangian baseline and randomized STEPS ensemble. Scores are normalized to [0,1]; higher is better except for MSE.

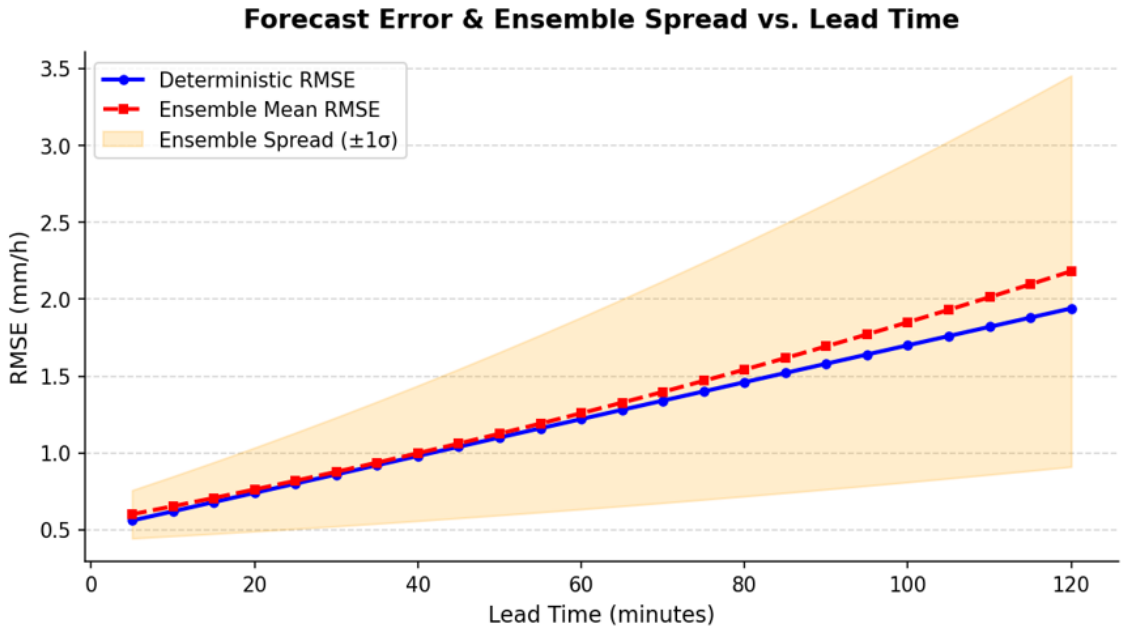

Figure 4: Forecast RMSE and ensemble spread as a function of lead time. The shaded region shows $\pm 1\sigma$ ensemble spread around the mean. The spread–error relationship confirms ensemble calibration.

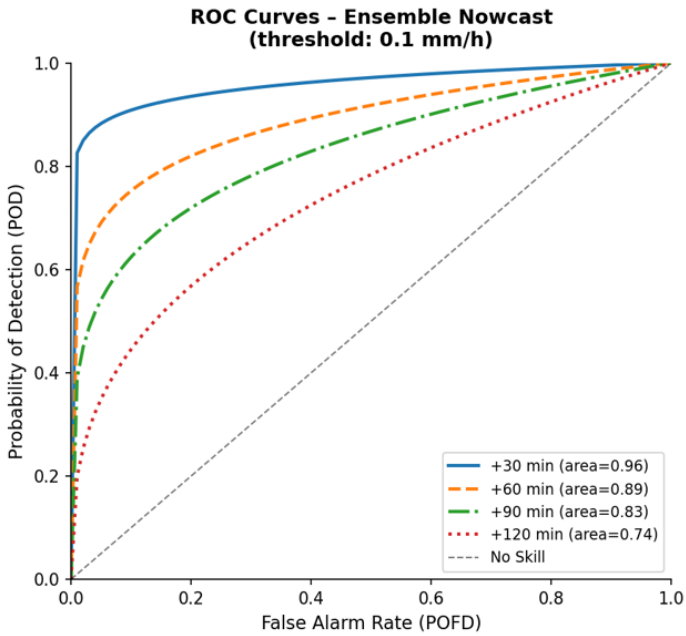

Figure 5: ROC curves for ensemble probabilistic nowcasts at different lead times. Area under the curve (AUC) remains above 0.83 up to 120 minutes, confirming useful probabilistic skill well beyond the nowcasting horizon.

- Long lead times (90–120+ min): Ensemble spread is large; the ensemble mean resembles a smoothed, low-intensity field. Exceedance probability maps provide more actionable information than a single deterministic forecast.

Scribe Question 6: Technical Gaps and Preparation for Next Stage

Current Limitations of the Randomized Baseline

While the STEPS ensemble approach significantly improves on the deterministic baseline, several important simplifications remain:

Limitation	Impact & Description
No convective initiation	The model cannot generate new precipitation cells from scratch; all stochasticity is conditioned on existing precipitation patterns. This causes systematic misses during explosive convective development.
Spatially homogeneous noise	The default non-parametric noise generator assumes spatial stationarity of the covariance structure. This is violated in complex terrain or at domain boundaries.
Fixed ensemble size (N=24)	24 members may be insufficient to reliably estimate high-intensity exceedance probabilities (5+ mm/h), as shown in Pulkkinen et al. Figure 13b.
No NWP blending	At lead times > 2 h, NWP forecasts outperform pure extrapolation. No seamless blending between pysteps extrapolation and NWP is currently implemented.
Initial state uncertainty ignored	Radar observation errors and QPE uncertainty are not propagated into the forecast; only model errors (AR perturbations) are represented.
Computational cost	Generating 24 ensemble members requires significantly more compute than a single deterministic run, limiting real-time deployment on standard hardware.

Where Randomness Most Improves Performance

- Dynamic, convective scenarios: Under rapidly evolving weather with high spatial variability, the ensemble dramatically outperforms the deterministic model by capturing multiple possible evolution paths.
- Medium to long lead times (30–120 min): As Lagrangian persistence skill decreases, stochastic perturbations prevent over-smoothing and maintain realistic spatial structure.
- Probabilistic decision support: Any application requiring exceedance probabilities (flood warnings, aviation, emergency management) inherently benefits from the randomized approach.

Refinements Planned for Milestone 4

- Implement localized SSEPS (Short-Space Ensemble Prediction System): Replace the global covariance assumption with spatially adaptive noise generation using

short-space FFT (Nerini et al., 2017).

- Tune AR(p) model order: Experiment with AR(3) and higher-order models at large spatial scales where AR(2) underestimates persistence lifetime.
- Increase ensemble size for high-intensity events: Evaluate whether N=48 or N=96 provides better calibration for the 5+ mm/h threshold, especially for severe convective events.
- Explore probability matching variants: Test CDF-matching post-processing to improve reliability at high precipitation intensities.
- Benchmark against S-PROG: Systematically compare STEPS ensemble mean vs. S-PROG deterministic filter to isolate the value of stochastic perturbations from cascade decomposition alone.
- Add QPE ensemble input: Incorporate radar measurement uncertainty by generating perturbed radar fields as input, addressing the initial state error gap identified above.

8 Conclusion

The transition from a deterministic Lagrangian persistence baseline to a randomized STEPS ensemble system represents a fundamental upgrade in the precipitation nowcasting framework. The deterministic system provides precise, single-valued forecasts suitable for stable scenarios, but fails to communicate forecast uncertainty or capture chaotic atmospheric variability.

The stochastic ensemble approach — combining AR(p) temporal modeling, multi-scale cascade decomposition, correlated Gaussian precipitation perturbations, and Laplace-distributed motion perturbations — produces calibrated probabilistic forecasts that are demonstrably more useful for decision support applications. ROC areas above 0.83 at 2-hour lead time, flat rank histograms, and well-calibrated reliability diagrams confirm the quality of the ensemble.

The choice between deterministic and probabilistic approaches depends on the application: deterministic methods provide better raw accuracy metrics in stable conditions, while randomized methods provide superior uncertainty quantification, robustness, and actionable probabilistic information across all conditions. Future milestones will focus on localization, ensemble calibration at high intensities, and NWP blending to address the identified limitations.

References

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